

Second Terminal Examination - 2079

Class : XII

Subject: Chemistry (3021)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- If the concentration of reactants is increased by 'x' then the k becomes
 a. $\ln \frac{k}{x}$ b. $\frac{k}{x}$ c. $k + x$ d. k
- What is the concentration of nitrate ions if equal volume of 1M NaNO₃ and 1M KCl are mixed?
 a. 0.25M b. 0.5M c. 1M d. 2M
- Which one is the strongest Bronsted base in the following ions?
 a. ClO⁻ b. ClO₂⁻ c. ClO₃⁻ d. ClO₄⁻
- The reduction potential of Al and Mg are -1.66 and -2.37 volts respectively. The Mg in chemical reaction,
 a. will be replaced by Al b. will replace Al
 c. will not able to replace Al d. None
- Bell metal is an alloy of
 a. Cu, Pb and Sn b. Sn and Cu
 c. Zn and Pb d. Zn, Cu and Sn

6. AgNO_3 does not give precipitate with chloroform because
- CHCl_3 is insoluble in water
 - CHCl_3 does not ionize in water
 - CHCl_3 is an organic compound
 - AgNO_3 is insoluble in CHCl_3
7. When ethyl alcohol is treated with acidified potassium dichromate it forms acetaldehyde. It is an example of
- Molecular rearrangement
 - Hydrolysis
 - Oxidation
 - Reduction
8. The reaction of benzene with Haloalkane in presence of anhydrous AlCl_3 is
- Friedel craft's reaction
 - Wurtz's reaction
 - Frankland reaction
 - Reimer Tiemann's reaction
9. Which of the following reagents cannot be used to distinguish between phenol and benzyl alcohol ?
- FeCl_3
 - NaHCO_3
 - Br_2 water
 - NaOH
10. The reagent that reacts with nitromethane to form methylhydroxylamine is
- Zn / HCl
 - $\text{Zn} / \text{NH}_4\text{Cl}$
 - Zn / NaOH
 - Sn / HCl
11. Ethers on hydrolysis give
- Carboxylic acid
 - Alcohols
 - Esters
 - Ketones

Group - B

Give short answer to the following questions.

[8 × 5 = 40]

1. Oxalic acid crystal is a primary standard substance.
 - a. What are the prerequisite for a substance to be primary standard? [2]
 - b. Convert the following: [2]
 - i. 2.5 M H_3PO_4 into normality
 - ii. 4.9 M H_2SO_4 into g/L.
 - c. Calculate the mass of oxalic acid, $(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$ required to prepare 100 mL of its decinormal solution. [1]
2. J.N. Bronsted and T.M Lowry put forward a new concept of acid and base. This concept is much broader than Arrhenius theory of acid and bases.
 - a. Why is Bronsted-Lowry theory superior to Arrhenius theory? [1]
 - b. Bronsted theory explains the formation of conjugate acid and base. What are conjugate acid and base pair? Give conjugate acid or conjugate base of HCl and SO_4^{--} [2]
 - c. Is H_2O acid or base according to Bronsted-Lowry theory? Justify [2]
3. Order of reaction is valuable tool for determining mechanism of reactions.
 - a. Derive integrated differential rate equation for first order reaction. [3]
 - b. From the following data, determine the overall order of the reaction and the order with respect to A. [2]

Initial concentration of A(M)	Initial rate(M/Sec)
0.05	3×10^{-5}
0.10	12×10^{-5}
0.20	48×10^{-5}

OR

The figure given below is calomel electrode. This can be used as an alternative for SHE. Both calomel electrode and SHE are reference electrodes. The reduction potential of calomel electrode depends upon the concentration of KCl solution used.



- a. Define standard electrode potential. [1]
 - b. Draw the calomel electrode showing its pertinent parts. Also write cell reactions when it acts as anode and cathode. [2]
 - c. Which of these two electrodes is primary and secondary reference electrode? Why are they so called? [2]
4. a. Tetra amine zinc sulphate is a complex compound. Write the following for the compound. [2]
- | | |
|--------------------------------|-------------------------|
| i. Charge on central metal ion | ii. Ligand |
| iii. Shape of complex ion | iv. Coordination number |
- b. What is d-d transition? [1]
 - c. Compounds of V^{5+} are colourless and V^{3+} are coloured. Give reason. [1]
 - d. Write molecular formula and one use of lunar caustic. [1]

5. Zinc and Mercury are volatile metals which belong to group IIB in the periodic table.
- Write all chemical reactions taking place in the extraction of zinc from its ore. [2]
 - Write the action of heat on white vitriol. [1]
 - How does calomel and corrosive sublimate react with ammonia? [2]
6. Give reason:
- OH group in phenol is ortho and para director towards electrophilic substitution reaction. [2]
 - Trichloromethane is stored in dark brown bottle containing 1% ethanol. [2]
 - Alcohols are less acidic than water. [1]
7. Pick the suitable reagents from column 'A' for the name of reaction in column 'B'. Give an example of each. [1+4]

Column A	Column B
Reagents	Name of Reactions
Na / ether , heat	Williamson's synthesis
Cu_2Cl_2 / HCl	Dehydrohalogenation
CH_3ONa	Sandmeyer's reaction
alc. KOH	Reimer Tiemann's reaction
$\text{NaOH} + \text{I}_2$	
CHCl_3 / KOH	
Cu/HCl	
$\text{CH}_3\text{Cl} / \text{AlCl}_3$	

8. Ethers have wide use as commercial solvents and extractants for esters, gums, hydrocarbons, alkaloids, oils, resins, dyes, plastics, lacquers, and paints.
- How is anisole prepared from phenol?
 - How is ethoxyethane converted to methoxyethane?
 - Write the action of ethoxyethane on:
 - Excess HI
 - conc. H_2SO_4 (cold)
- [1+2+2]

OR

Benzene is the “first and simplest” aromatic compound. There are many monosubstituted derivatives of benzene like chlorobenzene, phenol, nitrobenzene etc.

- a. Write the action of phenol on;
 - i. aq. Br_2
 - ii. Mixture of conc. HNO_3 and conc. H_2SO_4 ?
 - iii. Benzenediazonium chloride?
- b. Starting from nitrobenzene how would you prepare?
 - i. p – aminophenol
 - ii. Hydrazobenzene

[3+2]

Group - C

Give long answer to the following questions.

[3 × 8 = 24]

9. a. Define:
- i. Basicity of acid
 - ii. Decinormal solution
- b. Write differences between primary standard solution and secondary standard solution.
- c. 0.18 gram of a metal was completely dissolved in 100 cc of $\frac{N}{2}$ HCl solution. The whole volume was then made up to 500 cc. 25 cc of this diluted acid required 17.5 cc of $\frac{N}{10}$ alkali for complete neutralization. Find the equivalent weight of the metal.

[2+2+4]

OR

- a. A first order reaction was started with a decimolar solution of the reactant. After 8 minutes and 20 seconds, its concentration was found to be $\frac{M}{100}$.
- i. What is first order reaction? Write an example. [1]
 - ii. Determine the rate constant of the reaction. [1]
 - iii. What time would it require to complete half of the reaction? [2]

b. A part of electrochemical series is shown below:

Electrode	Standard reduction potential (E° Volt)
Li^+/Li	-3.05
K^+/K	-2.925
Ca^{++}/Ca	-2.870
Na^+/Na	-2.714
Mg^{++}/Mg	-2.370
Zn^{++}/Zn	-0.76
Fe^{++}/Fe	-0.44
Co^{++}/Co	-0.28
Ni^{++}/Ni	-0.25
Pb^{++}/Pb	-0.126
Fe^{3+}/Fe	-0.036
H^+/H_2	0
$\text{Sn}^{4+}/\text{Sn}^{2+}$	+0.15V
Cu^{++}/Cu	+0.34V
I_2/I^-	+0.524
$\text{Fe}^{3+}/\text{Fe}^{2+}$	+0.77

i. Write cell notation for the galvanic cell formed by using Mg^{++}/Mg and Fe^{++}/Fe electrode. Which one is anode and which one is cathode? [2]

ii. Name any two metals that can displace hydrogen from 1M HCl. [1]

iii. Predict whether the following reactions occur spontaneously or not?



10. a. How is blister copper obtained from copper matte? [2]

b. How does blue vitriol react with potassium iodide solution? [1]

c. Write the reactions involved in the different zones of Blast furnace in the extraction of iron. [2]

d. Write short note on manufacture of steel by Open Hearth process. [3]

11. Ethanol is the most common primary alcohol having various industrial applications like solvent, in manufacture of pharmaceuticals alcoholic beverages etc.

- a. How is ethanol prepared from
 - i. Ethene by oxo-process?
 - ii. Sucrose by fermentation?
- b. How is ethanol converted into
 - i. Propan-1-ol
 - ii. Ethanal
 - iii. Nitrolic acid
 - iv. Ethyl acetate
- c. Write the difference between distilled and undistilled alcoholic beverage. [3+4+1]

OR

- a. A haloalkane 'P' reacts with aq. KOH to produce 'Q'. 'Q' on oxidation with acidified $K_2Cr_2O_7$ produces 'R'. 'R' reacts with trichloromethane in presence of alkali produces hyponitric acid. 'Q' and 'R' both produce yellow precipitate with NaOH and I_2 with characteristic hospital smell. Identify P, Q and R. Also write the reactions involved.
- b. What happens when above compound 'P' is heated with Na/ dry ether?
- c. Write two methods applied to prepare compound 'Q'. [5+1+2]

☞ - The End - ☞